

Writing Two-Step Equations from Real-World Contexts

Additional Practice 7A.01.06 **ANSWER KEY**

For questions 1 through 7, write an equation, solve it, and write one sentence saying what your solution means.

1. A realtor earns a base salary of \$30,000 a year plus 3% of her yearly sales. Victoria wants her total earnings this year to be \$60,000. How much will she need to sell?

Equation. $30,000 + 0.03s = 60,000$

$$0.03s = 30,000$$

$$s = 1,000,000$$

Check: $30,000 + 0.03(1,000,000) = 30,000 + 30,000 = 60,000 \checkmark$

Meaning. *Victoria needs to sell \$1,000,000 worth of property this year.*

Her salary is \$30,000 and her commission is \$30,000. The \$60,000 is the two added together.

2. Fiona's favorite meal is on sale for 10% off the combo price. The sale is taken first, and then she uses a coupon for \$3 off. Fiona paid \$8.25 before taxes. What was the original price of the meal?

Equation. $0.90p - 3 = 8.25$

$$0.90p = 11.25$$

$$p = 12.50$$

Check: $0.90(12.50) = 11.25$, and $11.25 - 3 = 8.25 \checkmark$

Meaning. *The meal was originally priced at \$12.50.*

10% off means paying 90%. Multiply by 0.90, not 0.10.

3. A storage tank held 4,500 liters and is draining at 125 liters per hour. It now holds 2,750 liters. How long has it been draining?

Equation. $4,500 - 125h = 2,750$

$$-125h = -1,750$$

$$h = 14$$

Check: $4,500 - 125(14) = 4,500 - 1,750 = 2,750 \checkmark$

Meaning. *The tank has been draining for 14 hours.*

The rate is negative because the amount is going down.

4. Eight campers each bought a lunch and a bottle of water. A bottle of water costs \$4.25. The total for the eight came to \$102.00. What did one lunch cost?

Equation. $8(4.25 + n) = 102$

$$4.25 + n = 12.75$$

$$n = 8.50$$

Check: $8(4.25 + 8.50) = 8(12.75) = 102 \checkmark$

Meaning. *One lunch cost \$8.50.*

Dividing both sides by 8 first is quicker here than distributing.

5. A store adds \$1.60 in shipping to each of five books in an order. The order came to \$73.00. What is the price of one book?

Equation. $5(b + 1.60) = 73$

$$b + 1.60 = 14.60$$

$$b = 13.00$$

Check: $5(13 + 1.60) = 5(14.60) = 73 \checkmark$

Meaning. *One book costs \$13.00.*

$5b + 1.60 = 73$ charges shipping once instead of five times, and gives \$14.28 — close enough to look plausible, which is why the parentheses matter.

6. A jacket is marked 30% off. After the discount is taken, Marisol uses a \$5 coupon and pays \$47.50. What was the original price of the jacket?

Equation. $0.70p - 5 = 47.50$

$$0.70p = 52.50$$

$$p = 75.00$$

Check: $0.70(75) = 52.50$, and $52.50 - 5 = 47.50 \checkmark$

Meaning. *The jacket was originally priced at \$75.00.*

30% off means paying 70%.

7. Devon earns \$480 a week plus a 6% commission on his sales. Last week his total earnings were \$792. How much did he sell?

Equation. $480 + 0.06s = 792$

$$0.06s = 312$$

$$s = 5,200$$

Check: $480 + 0.06(5,200) = 480 + 312 = 792 \checkmark$

Meaning. *Devon sold \$5,200 worth of product last week.*

His commission was \$312, not \$792. The \$792 is his total earnings.

Question 8 is error analysis.

8. Mr. Humphries told his class he has some quarters and four dimes in his pocket, worth \$2.40 in all. Devin wrote $25x + 40 = 2.40$. Explain Devin's error, then write two different equations that are both correct.

Devin mixed his units. He wrote the coins in cents (25 and 40) but the total in dollars (2.40).

Correct in dollars: $0.25x + 0.40 = 2.40$

Correct in cents: $25x + 40 = 240$

Either one gives $x = 8$. Mr. Humphries has 8 quarters.

Both equations are correct and they are the only two that are. This is not an “answers will vary” question — a student who writes $0.25x + 40 = 2.40$ has made the same error Devin did, just in the other direction.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. In each one the unknown appears on both sides of the equal sign.

9. One moving company charges \$90 plus \$40 an hour. Another charges \$150 plus \$28 an hour. After how many hours do the two companies cost the same?

Equation. $90 + 40h = 150 + 28h$

$90 + 12h = 150$

$12h = 60$

$h = 5$

Check: $90 + 40(5) = 290$, and $150 + 28(5) = 290$ ✓

Meaning. *At 5 hours both companies charge \$290.*

For jobs shorter than 5 hours the first company is cheaper; after that, the second one is.

10. Sara has \$260 and saves \$15 a week. Theo has \$500 and spends \$25 a week. After how many weeks will they have the same amount of money?

Equation. $260 + 15w = 500 - 25w$

$260 + 40w = 500$

$40w = 240$

$w = 6$

Check: $260 + 15(6) = 350$, and $500 - 25(6) = 350$ ✓

Meaning. *After 6 weeks Sara and Theo will each have \$350.*

Adding $25w$ to both sides is what turns $-25w$ into a positive term. Sara is going up and Theo is going down, so they must meet.